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INGENIARE STRUCTURAL ENGINEERING CONSULTANCY

Structural Assessment Scope Guide

What to include when requesting structural integrity evaluation for an existing building

Prepared for building owners, administrators, facility managers, and decision-makers

Strength Begins With Certainty

Use this guide before requesting quotations, or comparing proposals for structural evaluation services.



Why this guide matters

Requesting a structural integrity evaluation is not the same as asking for a simple site visit. For existing buildings, the proper scope depends on the building age, use, visible condition, available records, occupancy, previous renovations, exposure, and the purpose of the assessment.

A clear scope helps prevent incomplete findings, unclear recommendations, and proposals that are difficult to compare. It also helps decision-makers understand whether the assessment should be limited to initial screening or should include testing, analysis, and engineering recommendations.

When to consider structural evaluation

- The building is more than 10 to 15 years old, especially if used by the public or occupied by many people.
- There are visible cracks, deflection, settlement, corrosion, water intrusion, spalling, or excessive deterioration.
- The building has undergone renovations, change of use, added loads, or undocumented alterations.
- The building experienced earthquakes, typhoons, flooding, fire, impact, or other events that may affect structural capacity.
- The owner, administrator, board needs a professional basis for repair, retrofitting, due diligence, occupancy, or capital planning.

Recommended information to prepare before requesting a proposal

Information	Why it matters
Building location and use	Helps define exposure, occupancy, risk, and applicable assessment approach.
Approximate floor area and number of storeys	Helps estimate inspection effort, testing quantity, analysis scope, and professional fee.
Estimated age of structure	Helps determine whether deterioration, code changes, and material verification should be considered.
Available structural or architectural plans	Allows comparison between intended design, actual configuration, and observed site condition.
Photos of visible concerns	Helps identify whether the concern is localized, widespread, urgent, or requires testing.
Reason for the assessment	Clarifies whether the goal is screening, repair planning, due diligence, retrofit design, or compliance.

Typical structural assessment scope items

1. Document review

- Review of available structural, architectural, as-built, renovation, or maintenance records.
- Initial identification of missing information, drawing inconsistencies, and areas requiring field verification.

2. Visual inspection

- Observation of structural elements such as columns, beams, slabs, walls, roof framing, foundations where accessible, and visible connections.
- Photo documentation of cracks, deflection, settlement, corrosion, spalling, water damage, deterioration, or unsafe conditions.

3. Non-destructive testing and scanning, when required

- Rebound hammer testing may be used for preliminary concrete surface hardness assessment and uniformity checks.
- Rebar scanning or GPR may be used to locate reinforcement, estimate spacing, and verify embedded elements where applicable.
- Additional testing may be recommended if visual findings indicate material uncertainty or concealed defects.

4. Material verification, when required

- Concrete coring and laboratory compression testing may be required when actual concrete strength must be verified.
- Rebar extraction or other tests may be considered only when necessary and acceptable to the owner because they are invasive.

5. Structural analysis

- Analysis may be needed when the assessment requires capacity verification, evaluation of added loads, seismic or wind demand checks, or retrofit planning.
- The model and assumptions should be based on available drawings, field measurements, and test results as applicable.

6. Structural condition report

- The report should clearly summarize observations, test results if any, engineering interpretation, limitations, conclusions, and recommended next steps.
- When appropriate, the report should identify probable contributing factors such as design issue, construction execution issue, material issue, loading issue, site deviation, deterioration, or a combination of factors.

Scope checklist for request for quotation

- ☐ Project background and purpose of assessment
- ☐ Building age, use, floor area, number of storeys, and location
- ☐ Site inspection and visual condition assessment
- ☐ Photo documentation and marked-up affected areas
- ☐ Review of available plans and existing records
- ☐ Field verification of structural layout and member sizes
- ☐ Non-destructive testing requirements, if applicable
- ☐ GPR or rebar scanning requirements, if applicable
- ☐ Concrete coring and laboratory testing, if applicable
- ☐ Structural analysis requirements, if applicable
- ☐ Interim safety memo, if urgent conditions must be communicated early
- ☐ Final signed and sealed structural condition report
- ☐ Repair, strengthening, or retrofitting recommendations, if included
- ☐ Exclusions, assumptions, limitations, and client responsibilities

Important note: visual inspection alone may not be enough

Visual inspection is a useful first step, but it cannot confirm all hidden conditions. Some concerns may require scanning, testing, structural analysis, or further investigation. The appropriate scope should be selected based on the risk, building condition, available records, and intended use of the results.

Suggested evaluation levels

Level	Suitable for	Typical inclusions
Initial Screening	Early concern, budget planning, or preliminary due diligence	Visual inspection, photo documentation, initial observations, recommended next steps
Detailed Assessment	Older buildings, visible distress, public occupancy planning	Document review, inspection, selected NDT/scanning/testing, engineering evaluation, condition report
Analysis and Retrofit Planning	Buildings requiring capacity verification, strengthening, or change of use	Detailed assessment, structural analysis, retrofit recommendations, possible repair or strengthening design as a separate phase

How to compare proposals

- Check whether all consultants are pricing the same scope. A lower fee may mean fewer inspection items, fewer tests, no analysis, or limited deliverables.
- Review the deliverables. A professional assessment should clearly state what will be submitted, whether signed and sealed, and how many sets or digital files will be provided.
- Clarify exclusions. Some items such as coring, laboratory testing, GPR, access equipment, scaffolding, destructive investigation, or retrofitting design may be excluded or optional.
- Ask whether the scope is fit for the intended decision. A simple screening report may not be enough for retrofit design, major renovation, or formal decision-making.

Next step

If you are preparing a request for structural integrity evaluation, start by listing the building details, available documents, visible concerns, and intended purpose of the assessment. A professional engineer can then recommend whether the scope should be limited, staged, or expanded based on the actual condition and risk of the structure.

Ingeniare Structural Engineering Consultancy

Structural Integrity Evaluation | Structural Analysis | Retrofitting Recommendations | Engineering Due Diligence

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